GG-G-531C
June 18, 1978
SUPERSEDING
GG-G-531B
Apri1 9, 1967

FEDERAL SPECIFICATION

GOGGLES, INDUSTRIAL AND SPECTACLES, INDUSTRIAL, PLASTIC

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1, 1 Scope. This specification covers industrial, eye-protective goggles and spectacles which are constructed primarily of plastic materials,

1.2 Classification.

1.2. 1 Types, grades, classes, styles, and designs. Goggles and spectacles covered by this specification shall be of the following types, grades, classes, styles, and designs, as specified (see 6. 2).

Type I - Goggles.

Grade A - Anti-fog treated lens.

Grade B - Untreated lens.

Class 1 - Clear lens.

Class 2 - Antiglare lens.

Type II - Spectacles,

Style a - With side shields

Design H - Center hinged bridge.

Design R - Unhinged, rigid bridge.

Style b - Without side shields

Class 1 - Clear lenses.

Class 2 - Antiglare lenses,

2. APPLICABLE DOCUMENTS

2. 1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, forma part of this specification to the extent specified herein.

Federal Specification:

PPP-B-636 Boxes, Shipping, Fiberboard.

FSC 4240

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:51Z
Check the source to verify that this is the current version before use.

Federal Standard:

Fed. Std. No. 406 - Plastics: Methods of Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale *on* a subscription basis by the Superintendent of Documents, U, S, Government Printing 'Office, Washington, DC 20402,

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, Seattle, WA, and Philadelphia.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks, and the Index of Federal Specifications and Standards from established distribution points in their agencies,)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer,)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards Institute (ANSI) Standards:

ANSI Z87. 1 - Practice for Occupational and Educational Eye and Face Protection,

ANSI Z80. 3 - Requirements for Nonprescription Sunglasses and Fashion Eyewear.

(Application for copies should be addressed to the American National Standards Institute, Inc. , 1430 Broadway, New York, NY 10018.)

National Motor Freight Traffic Association, Inc. , Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, hc. , Traffic Department, 1616 P Street, N, W, , Washington, DC 20036,)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:51Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3. 1 Standard commercial product. Except as modified by the provisions of this specification, goggles and spectacles shall be the manufacturer's standard commercial product.

3.2 Materials. Lenses and frames of goggles and spectacles covered by this specification shall be of suitable plastics. Hinges, fasteners, and reinforcements may be fabricated of suitable metals.

3.2. 1 Disinfection, Materials used in goggles and spectacles shall show no sign of deterioration or discoloration when subjected to the disinfection procedure described in 4,4.

3. 2.2 Corrosion resistance. Metal parts shall show no signs of corrosion when tested in accordance with 4.4.

3. 2.3 Water absorption. Plastic parts shall show not more than 5 percent water absorption when tested in accordance with 4.4.

3. 2.4 Flammability. The rate of flame-propagation for plastic components shall be 24 seconds per inch or less when tested in accordance with 4.4,

3.3 Type I. Goggles shall consist of a plastic frame or mask, plastic lens, and adjustable headband of a suitable elastic material such as braided elastic, rubber strap, or other similar materials. Goggles shall be designed to generally contour to the shape of the face, covering completely the eye sockets and the facial area surrounding the eyes, offering both frontal and side protection to the wearer. Goggles shall be capable of being worn over conventional spectacles with no decrease in protection offered. The lens portion of the goggles shall be not less than 5-3/8 inches long (outside measurement) at the longest dimension and not less than 2-3 / 16 inches (outside measurement) at the widest dimension. Goggles shall be furnished with either integral or replaceable lens at the manufacturer!s option, unless otherwise specified (see 6. 2). Weight of complete goggles shall not exceed 3.0 ounces.

3.3, 1 Frame or mask. Frames or masks shall be clear, translucent, colored, or opaque, as specified (see 6. 2). They shall be sufficiently rigid to hold the lens securely in place before the wearer's eyes, Edges which contact the wearer's face shall be rounded. Ventilation openings shall be provided to permit circulation of air through the goggles. These openings shall not permit the passage of a sphere 5/64 inch in diameter. Methods of securing the lens in the frame or mask shall be sufficient to securely hold the lens in place when the fracture resistance test specified in 4, 4 is conducted.

3,3.2 Lenses, Lenses shall not chip, fracture, be pentrated, or shatter when subjected to the impact resistance test in 4.4.

3.3.2. 1 Grade A, Grade A lenses, when tested in accordance with 4, 5, shall show no visible fog, The anti-fogging treatment shall not be affected when used or cleaned according to the manufacturers directions.

3. 3.2.2 Class 1. Lenses of class 1 goggles shall be of clear plastic with a luminous transmittance of not less than 85 percent when tested in accordance with 4, 4. They shall be free from striae, waves, and other visible defects which might impair optical qualities. Lenses shall be not less than O. 050 inch thick,

3. 3.2,3 Class 2. Lenses of class 2 goggles shall be of antiglare design. They shall meet requirements for class 1 lenses except that they shall be of colored plastic capable of reducing harmless glare and absorbing ultraviolet radiation, Specifically, class 2 lenses shall be capable of meeting the color limits, traffic signal transmittance e, and ultraviolet transmittance requirements for lenses suitable for daylight driving as detailed below when tested in accordance with

(a) Color limits - values of X and Y shall not fall outside prescribed regions on the CIE (1931) standard chromaticity diagram in ANSI Z80. 3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:51Z
Check the source to verify that this is the current version before use.

(b) Traffic signal transmittance . 8 percent minimum for red, 6 percent minimum for yellow, 6 percent minimum for green.

(c) Ultraviolet transmittance - no greater than O. 2 percent,

3. 3.3 Headband. Headbands of suitable elastic material shall be not less than 7/16 inch wide. They shall be adjustable in length and shall be fastened to the goggle frame in such a manner as to be easily replaceable, They shall not interfere with the facial fit of the goggles. The headband shall withstand, without visible deterioration, the disinfection test in 4, 4,

3.4 Type II, Spectacles shall consist of a frame, lenses, nose bridge, and temples.

3.4. 1 Frames. Spectacle frames shall be all plastic and shall be clear, translucent, colored, or opaque, as specified (see 6. 2). The use of metal' in the temple or bridge hinges is allowed.

3.4. 1.1 Style a. Style a spectacles shall be furnished with side shields of clear or colored (antiglare) plastic, as specified (see 6. 2). Side shields shall be securely attached to temples and/or frames in such a manner so to provide full protection from side impact. Side shields shall extend from the plane of the lenses, along the temples for a distance of at least 1 3/4 inches, Antiglare side shields shall be capable of meeting the requirements for color limits, traffic signal transmittance, and ultraviolet transmittance expressed in 3, 3.2.3.

3.4.1. 1.1. Design H. Design H spectacles shall have a hinged bridge which enables them to open to fit broad faces and close to fit narrow ones , and to offer maximum protection to each.

3.4, 1.1.2. Design R. Design R spectacles shall have a rigid bridge, nonadjustable, and shall be contoured to the general curve of noses.

3, 4, 1.2 Style b. Style b spectacles shall be furnished without side shields, They shall be of rigid bridge design only.

3, 4, 2 Lenses. Lenses shall not chip, fracture, be penetrated, or shatter when subjected to the impact resistance test in 4,4.

3. 4.2.1 Class 1. Lenses of class 1 spectalces shall be of clear plastic with a luminous transmittance of not less than 85 percent when tested in accordance with 4, 4. They shall be free from striae, waves, and other visible defects which might impair optical qualities, Lenses shall be not less than O. 050 inch thick.

3. 4.2.2 Class 2. Lenses of class 2 goggles shall be of antiglare design. They shall meet require. ments for class 1 lenses except that they shall be of colored plastic capable of reducing harmless glare and absorbing ultraviolet radiation. They shall be capable of meeting the requirements for color limits, traffic signal transmittance, and ultraviolet transmittance expressed in 3.3.2.3.

3. 4.3 Temples. Spectacles shall be provided with spatula type temples of the manufacturers commercial design, Temples shall be of all plastic or of metal-reinforced plastic at the manufacturers option.

3,4.4 Weight. Spectacles with side shields shall weigh not more than 1 3/4 ounces and spectacles without side shields shall weigh not more than 1 1 /2 ounces.

3.5 Identification marking. All goggles and spectacles furnished under this specification shall be permanently and legibly marked with the manufacturers name or recognized trademark.

3.6 Workmanship, Workmanship shall be of the quality prevailing among manufacturers normally producing goggles and spectacles of the types specified herein. All goggles and spectacles furnished shall be free from any defects which might affect appearance, functionally, or serviceability.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:51Z
Check the source to verify that this is the current version before use.

QUALITY ASSURANCE PROVISIONS

4, 1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified, Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government.

4, 2 Sampling.

4.2, 1 Sampling for examination. A random sample of goggles or spectacles shall be selected from each lot in accordance with MI L- ST D- 105, inspection level II. The acceptable quality level (AQL) shall be 2.5 defects per 100 units.

4, 2, 2 Sampling for tests. Two representative pairs of goggles or spectacles, as applicable, shall be selected at random from each lot offered for testing in accordance with 4.4. Failure of either of the samples to pass any of the tests shall be cause for rejection of the lot.

4. 2.3 Sampling for fogging test, grade A goggles. Samples shall be selected from each lot of grade A goggles in accordance with level S-1 of MI L- STD-105. The acceptable quality level (AQL) for this test shall be 4. 0 defects per 100 units, when tested in accordance with 4. 5,

4. 3 Examination. Each of the samples selected in accordance with 4.2.1 shall be examined for the defects listed below. Finding of defects that exceed the AQL in number shall be cause for rejection of the lot represented by the samples. In no case shall a sample in which a defect has been noted be offered for delivery.

Defects:

(a) Goggles or spectacles not of type, grade, class, style, and design specified.

(b) Materials not as specified.

(c) Design and construction not as specified.

(d) Dimensions not within specified limits.

(e) Weight of goggles or spectacles greater than specified.

(f) Presence in lenses of striae, waves, or other visible defects.

(g) Goggle headbands less than specified width.

(h) Lack of proper identification marking.

(i) Defects which affect appearance, functionally, or serviceability.

4.4 Tests. Samples selected in accordance with 4.2, 2 shall be tested in accordance with the table below:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:51Z
Check the source to verify that this is the current version before use.

TESTS

Property	Requirement Paragraph	Test in accordance with
Disinfection	3. 2.1	ANZ1 Z87, 1 - Materials and Methods of Test of Protectors
Corrosion resistance	3. 2.2	ANZ1 287. 1 - Materials and Methods of Test of Protectors
Water absorption	3. 2.3	Fed. Test Method Std. 406 - Test Method 7031
Flammability	3. 2.4	ANZ1 Z87, 1 - Materials and Methods of Test of Protectors
Impact resistance	3. 3.2&3.4.2	ANZ1 Z87. 1 - Lenses
Luminous transmittance e	3.3. 2.2& 3,4.2. l	Fed, Test Method Std. 406 - Test Method 3022
Go lo r limits	3.3.2.3, 3.4, 1.1 & 3. 4.2.2	ANZ1 Z80. 3
Traffic signal transmittance	3.3.2,3, 3.4. 1.1 & 3,4. 2.2	ANZ1 Z80.3
Ultraviolet transmittance	3.3.2.3, 3.4. 1.1 & 3,4. 2.2	ANZ1 Z80. 3

4. 5 Fogging test, grade A goggles, Samples selected in accordance with 4.2, 3 shall be placed in a chamber and maintained at a temperature of 45 degrees F or less for a period of at least 1 hour. They shall *then* be removed and exposed to room conditions of 65-70 degrees F with a relative humidity of not less than 40 percent, Any visible fog forming on the lens of a sample shall constitute failure of the test, If the number of sample failing this test exceeds the stated AQL, the lot represented by these samples shall be rejected,

4.6 Examination of preparation for delivery. An examination shall be made to determine compliance with the requirements. of Section 5. The sample unit shall be one shipping container fully prepared for delivery, Sampling shall be in accordance with MIL-STD- 105. The inspection level shall be S-2 with an AQL of 4, 0 expressed in terms of percent defective,

5, PREPARATION FOR DELIVERY

5, 1 Packaging. Packaging shall be level A or commercial, as specified (see 6. 2).

5, 1, 1 Level A. Each pair of goggles or spectacles shall be wrapped in non-abrasive tissue paper and packaged in a close-fitting box conforming to PPP-B-636, class domestic, style optional, Closure of the boxes shall be in accordance with the appendix to PPP. B-636,

5. 1.2 Commercial. Each pair of goggles or spectacle shall be packaged in accordance with normal commercial practice, The complete package shall be designed to protect the goggles or spectacles against damage during shipment, handling and storage.

5, 2 Packing. Packing shall be level A or commercial, as specified (see 6. 2),

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:51Z
Check the source to verify that this is the current version before use.

5.2, 1 Level A, Goggles or spectacles of like stock number, packaged as specified in 5. 1. 1, shall be packed in quantities as specified in the contract or order, in a close fitting box conforming to PPP-B-636, class weather resistant, style RSC or FOL. Closure and waterproof sealing of the boxes shall be in accordance with the appendix to PPP- B-636, Alternatively, cleated plywood, wire bound or nailed wood boxes shall be acceptable shipping containers when lined with a waterproof barrier material. The edges of the barrier material shall be sealed with waterproof tape *or* adhesive.

5, 2, 2 Commercial The goggles or spectacles of like stock number, packaged as specified in 5, 1.2 shall be packed in fiberboard boxes to insure delivery at destination, to provide for redistribution by the initial receiving activity, and shall be acceptable by common carrier under National Motor Freight Classification or Uniform Freight Classification.

5.3 Unitization, When shipments to Government depots are full car or truckload, the shipping containers shall be unitized to facilitate handling in accordance with normal commercial practice. The unitized load shall not exceed 2, 500 pounds in weight, 63 inches in height, 56 inches in length, and 45 inches in width.

5.4 Marking Packages, shipping containers, and unitized loads (when applicable) shall be marked as specified in the contract or order.

6. NOTES

6, 1 Intended use. Goggles and spectacles covered by this specification are intended to provide protection for eyes from hazards such as small flying particles of *dust,* chips, and machine cutting Type II, style B spectacles provide *frontal* protection only.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents :

(a) Title, number, and date of this specification.
(b) Type, grade, class, style, and design required (see 1.2. 1).
(c) If integral or replaceable lens is required (see 3. 3).
(d) Whether clear, translucent, colored, or opaque. frames are required (see 3. 3.1 and 3.4. 1).
(e) Whether clear *or* colored sideshields are required (see 3.4.1. l)'.
(f) Level of packaging and packing required (see 5. 1 and 5. 2),

Military custodians:

Army - EA
Air Force - 99

Review activities:

Army - EA
Air Force - 99, 11

User activity:

Army - ME

Preparing activity:

GSA-FSS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 60 cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:51Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:51Z
Check the source to verify that this is the current version before use.